A disproof of Conjecture 00000000806

The Dirichlet spectral gap ratio of two equal disks

1 The conjecture and the reading

Definition: The spectral gap $\gamma(\Omega) = \lambda_2(\Omega) - \lambda_1(\Omega)$ and its ratio to the first eigenvalue $R(\Omega) = \gamma/\lambda_1$. *Conjecture:* $\sup R(\Omega)$ is attained by the union of two equal-volume disjoint disks, with value $R = 3/4$; every connected domain has R strictly smaller.

The text does not name the eigenvalue problem. We read $\lambda_k(\Omega)$ as the k-th Dirichlet eigenvalue of the Laplacian on an open set $\Omega \subset \mathbb{R}^2$ of finite area, counted with multiplicity (the standard convention). The conjecture is read as the conjunction of

(i) some pair of disjoint open disks of equal radius $r > 0$ has $R(D_1 \cup D_2) = 3/4$;

(ii) $R(\Omega) \le 3/4$ for every nonempty open Ω of finite area;

(iii) $R(\Omega) < 3/4$ for every connected open Ω of finite area.

(Equal volume of two disks means equal radius.) We refute clause (i), hence the conjunction. Lean: `C806.Conjecture` is exactly this conjunction and `C806.not_conjecture : ¬ Conjecture`.

2 Mathematical content

Let D_1, D_2 be disjoint open disks of equal radius. A Dirichlet eigenfunction on D_1 extended by zero is an eigenfunction on $D_1 \cup D_2$, and likewise for D_2, with the same eigenvalue because D_2 is a translate of D_1. So the first eigenvalue of $D_1 \cup D_2$ is attained twice: $\lambda_1 = \lambda_2$, $\gamma = 0$ and $R = 0 \neq 3/4$.

3 The formalization (file `lean/Conjecture806/Basic.lean`, namespace `C806`)

We do not use a stipulated value of the eigenvalues. The Dirichlet eigenvalues are defined by the Courant–Fischer (min–max) variational principle, which is the standard definition of $\lambda_k(\Omega)$ for an open set of finite measure and equals the k-th eigenvalue of the Dirichlet Laplacian counted with multiplicity.

Definition 1 (as in Lean)**.** $E = \mathbb{R}^2$ (`EuclideanSpace ℝ (Fin 2)`) with Lebesgue measure. `testSpace Ω` is the real vector space $C_c^\infty(\Omega)$: functions $u : E \to \mathbb{R}$ with `ContDiff ℝ ∞ u` (smoothness order $(\top : \mathbb{N}_\infty)$, not analytic), compact support, and $\operatorname{tsupport} u \subseteq \Omega$. With values in $[0,\infty]$: $N(u) = \int^- \|Du\|^2$, $D(u) = \int^- |u|^2$, $Q(u) = N(u)/D(u)$ (the Rayleigh quotient), and

$$\lambda_k(\Omega) = \inf_{W \le C_c^\infty(\Omega),\ \dim W = k} \ \sup_{u \in W,\, u \neq 0} Q(u), \qquad R(\Omega) = \frac{\lambda_2(\Omega) - \lambda_1(\Omega)}{\lambda_1(\Omega)}$$

(`toReal` is applied to λ_k in the quotient; see conventions).

Source for the definition. Retrieved from `https://en.wikipedia.org/wiki/Dirichlet_eigenvalue`: "where each eigenvalue is counted according to its geometric multiplicity" (so $\lambda_1 \le \lambda_2$ with repetition) and "the infimum is taken over all u of compact support that do not vanish identically in Ω. By a density argument, this infimum agrees with that taken over nonzero $u \in H_0^1$". The same page describes the max–min principle for λ_k; the Courant–Fischer–Weyl min–max principle is described at `https://en.wikipedia.org/wiki/Min-max_theorem`. Because $C_c^\infty(\Omega)$ is dense in $H_0^1(\Omega)$, the min–max over C_c^∞ subspaces equals the min–max over H_0^1 subspaces; this density step is the one standard fact used in identifying our `lam` with the spectral object, and it is not reproved in Lean.

Non-vacuity. `testSpace_ne_bot` shows that every ball $B(c, r)$, $r > 0$, carries a nonzero element of C_c^∞ (a Mathlib `ContDiffBump`), so the quotients are not taken over an empty or trivial space.

4 Proof of $\lambda_1 = \lambda_2$ on $A \cup B$

Let $A = B(c_1, r)$, $B = B(c_2, r)$ be disjoint, $v = c_2 - c_1$; so $x \in B \iff x - v \in A$.

1. $\lambda_1 \le \lambda_2$ for every Ω (`lam1_le_lam2`). A 2-dimensional W contains $u \neq 0$; $\mathrm{span}\{u\}$ is 1-dimensional and $Q(cu) = Q(u)$ for $c \neq 0$ (`Q_smul`), so $\lambda_1 \le Q(u) \le \sup_W Q$.

2. *Pair.* If $p \in C_c^\infty(A)$, $p \neq 0$, let $q = p(\cdot - v) \in C_c^\infty(B)$. Then p, q have disjoint supports, so $|\nabla(ap+bq)|^2 = a^2|\nabla p|^2 + b^2|\nabla q|^2$ and likewise for $|ap+bq|^2$ (`N_comb`, `D_comb`); by translation invariance of Lebesgue measure $N(q) = N(p)$, $D(q) = D(p)$ (`N_tr`, `D_tr`). Hence Q is constant equal to $Q(p)$ on $\mathrm{span}\{p, q\} \setminus 0$ (`Q_pair`), and p, q are linearly independent (evaluate at a point where $p \neq 0$ and at its translate), so this span is an admissible 2-dimensional W in $\Omega = A \cup B$: $\lambda_2(A \cup B) \le Q(p)$ (`lam2_le_Q_A`). The same for $p \in C_c^\infty(B)$ via the inverse translation (`lam2_le_Q_B`).

3. *Splitting.* Let $u \in C_c^\infty(A \cup B)$, $u \neq 0$. Since A, B are open and disjoint and $\mathrm{tsupport}\, u \subset A \cup B$, the indicator pieces $u_A = 1_A u$, $u_B = 1_B u$ are smooth (locally equal to u or 0), lie in $C_c^\infty(A)$, $C_c^\infty(B)$ (`ind_mem`) and $u = u_A + u_B$ (`ind_add`). By (2), $\lambda_2\, D(u_\bullet) \le N(u_\bullet)$ for both pieces (trivially if a piece is 0); adding, $\lambda_2 D(u) \le N(u)$ using $N(u) = N(u_A) + N(u_B)$, $D(u) = D(u_A) + D(u_B)$. Since $0 < D(u) < \infty$ (continuous, compactly supported, nonzero; `D_ne_zero`, `D_ne_top`) this gives $\lambda_2 \le Q(u)$ (`lam2_le_Q`).

4. Therefore every 1-dimensional W has $\sup_W Q \ge \lambda_2$, i.e. $\lambda_2 \le \lambda_1$ (`lam2_le_lam1`). With (1), $\lambda_1 = \lambda_2$ (`R_two_disks`), so $R = (\lambda_2 - \lambda_1)/\lambda_1 = 0/\lambda_1 = 0$ in every case (also if λ_1 were 0 or ∞, since $x - x = 0$ and $0/y = 0$ in $\mathbb{R}$).

Theorem 2 (`R_two_disks`, `not_conjecture`)**.** *For all $c_1, c_2 \in \mathbb{R}^2$ and $r \in \mathbb{R}$ with $B(c_1, r) \cap B(c_2, r) = \emptyset$: $\lambda_1 = \lambda_2$ and $R(B(c_1, r) \cup B(c_2, r)) = 0$. Hence clause (i) fails and $\neg$Conjecture.*

5 Conventions and scope

- Eigenvalues are counted with multiplicity (Courant–Fischer). This is the reading refuted.

- The quotient is formed in $\mathbb{R}$ from `ENNReal.toReal` of λ_k, where $0/0 = 0$; the result $R = 0$ does not depend on the values of λ_k once $\lambda_1 = \lambda_2$.
- Quantification: the refutation covers *every* pair of disjoint open disks of equal radius (the existential clause (i) fails, a fortiori any universal version), not only one pair.
- **Not refuted in Lean.** A "distinct eigenvalues" convention (λ_2 = second *distinct* eigenvalue) is not formalized. Under it $R(D_1 \cup D_2)$ would equal the single-disk value $j_{1,1}^2/j_{0,1}^2 - 1 \approx 1.539$ (numerical computation, not a Lean statement), which is again not $3/4$; that remark is outside the formal proof.
- Clauses (ii) and (iii) are not used. (Numerically the single disk, which is connected, would have $R \approx 1.539 > 3/4$ in the distinct-eigenvalue reading, so (iii) looks false too; not formalized.)

6 Relation to the earlier submission

The earlier attempt (PR #1051, closed) was rejected because "the Lean project does not formalise the conjecture. The spectral gap and ratio are modelled as subtraction and a pair of natural numbers; no domain, Laplacian or eigenvalue is defined." This submission defines the domain (open subsets of $\mathbb{R}^2$), the Dirichlet energy and Rayleigh quotient on $C_c^\infty(\Omega)$, the eigenvalues $\lambda_k(\Omega)$ by the min–max principle, and R, and it proves $R(D_1 \cup D_2) = 0$ for *all* equal disjoint disks from these definitions.

7 Lean statement and axiom audit

Namespace `C806`: `testSpace`, `N`, `D`, `Q`, `lam`, `R`, `Conjecture`; theorems `testSpace_ne_bot`, `lam1_le_lam2`, `lam2_le_Q`, `lam2_le_lam1`, `R_two_disks`, `not_conjecture`. `#print axioms C806.not_conjecture` reports only `propext`, `Classical.choice`, `Quot.sound`; there is no `sorry` and no `native_decide`. File length: 417 lines.